Article

Proxy-Based Diagnostics of Quantum Oracle Sketching Robustness for Non-IID Sensor and Telemetry Streams

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Abstract: In theory, quantum oracle sketching achieves exponential memory advantages over classical streaming, and its guarantees extend to correlated non-IID data; how these guarantees behave under the structured correlation typical of sensor and telemetry streams, however, remains uncharted. We present a proxy-based diagnostic and hypothesis-generation framework for correlation-sensitive streaming analysis. Operational proxies for the refreshing time τ and repetition number r, inspired by the exact framework, are estimated on five synthetic non-IID regimes (IID, Markov switching, seasonal drift, burst repetition, long-range dependence) and mapped onto a (τ, r) sensitivity landscape—the paper's central artifact. Three memory-bounded classical baselines (online SGD, averaged SGD, Count-Min) supply empirical reference points; no quantum algorithm is implemented or simulated, and all quantum curves are heuristic proxy estimates. Analytic proxies overestimate the empirically measured refreshing time by up to 125× under long-range dependence (≈ 8× for Markov). Using empirical τ, the proxy model predicts a hypothesized quantum-favourable region under mild-to-moderate correlation that closes as correlation strengthens (mean-curve crossing at $\rho^* \approx 0.82$; by $\rho = 0.88$ the classical baseline exceeds the proxy estimate by 0.040 on average). Applied unchanged to two real NAB telemetry streams, the same estimators place NYC Taxi inside and Machine Temperature outside the hypothesized favourable region, with imbalance-aware metrics guarding against majority-class artefacts. Ablations over the τ estimator, forward window, stream length, target function, and proxy constants preserve the qualitative regime ordering.

Keywords: sensor data streams; telemetry; streaming algorithms; non-IID data; correlation diagnostics; quantum computing; oracle sketching

Citation: Farsi, M.; Mahmoud, M.; Helmy, A. Proxy-Based Diagnostics of Quantum Oracle Sketching for Non-IID Streams. *Sensors* **2026**, *1*, 0. https://doi.org/

Received:
Revised:
Accepted:
Published:

1. Introduction

The exponential growth of data-intensive applications in domains such as sensor networks, financial markets, climate monitoring, and telecommunications has created an urgent need for streaming algorithms that process data on the fly with severely constrained memory [37]. Classical streaming algorithms—including sketching methods, online learners, and sliding-window estimators—offer practical solutions but face fundamental memory–accuracy tradeoffs rooted in information-theoretic lower bounds [3]. A further complication is that real-world streams are rarely IID: Internet traffic exhibits long-range dependence and self-similarity [33,48], and streaming-ML systems must contend with concept drift and non-stationary input distributions [4,20,45,51].

Recent work by Zhao et al. [52] establishes, at the level of theory, exponential quantum advantages for classification and dimension reduction on massive classical data via quantum oracle sketching, using a polylogarithmic-size quantum computer ($\mathcal{O}(n)$ qubits for $n = \log_2 N$). For several concrete problems the proven classical memory requirement scales polynomially in $N = 2^n$, yielding an exponential theoretical gap; these classical hardness results are task-specific proofs, not empirical measurements [10,25,26,43]. The broader quantum ML landscape includes variational algorithms [13], NISQ-era perspectives [40], quantum learning theory [42], and classical shadow tomography [28].

A particularly notable feature of this framework is its treatment of non-IID data. The framework extends to correlated, non-IID data via a hierarchical data-generation model in which the quantum sample complexity is multiplied by the *repetition number* of the data-generation process. However, the practical implications of this robustness under structured, real-world correlation patterns remain unexplored.

Four questions motivate our study: (RQ1) How tight are the generic robustness guarantees for specific structured families? (RQ2) How informative are τ and r as accuracy predictors? (RQ3) Under what correlation conditions does the proxy-estimated favourable separation persist, degrade, or potentially improve? (RQ4) How do empirically measured (τ, r) relate to the analytic proxy predictions?

Scope

The scope is a **proxy-based computational study** of data-processing viability for sensor-style streams: operational correlation proxies inspired by this framework are used to map where proxy-estimated performance is most favourable under the modelling assumptions across structured synthetic non-IID conditions of the kinds documented for sensor and telemetry traffic (Markov switching, seasonal cycles, burst repetition, long-range dependence), with a two-dataset real-stream sanity check on NAB telemetry streams—an urban demand series and a machine-temperature sensor series (Section 6.8). Quantum performance curves are heuristic proxies and no quantum algorithm is implemented; circuit simulation, primary-source theorem reproduction, and broader real-stream validation remain explicit follow-on targets (Section 9). The paper is therefore best read as a **diagnostic and hypothesis-generation framework** for correlation-sensitive streaming analysis: it produces falsifiable, proxy-level predictions about where quantum oracle sketching is most likely to remain robust, and it neither demonstrates nor claims a realised quantum–classical separation.

We study five non-IID regimes (IID, Markov switching, seasonal drift, burst repetition, long-range dependence), specifying analytic proxy expressions for τ and r, measuring their empirical counterparts on generated streams, comparing three classical baselines, and mapping results onto a (τ, r) sensitivity landscape with 10-seed means, 95% CI bands, and paired comparisons interpreted through effect sizes and interval estimates.

Our principal contributions are:

1. **Operational correlation proxies** for five structured non-IID regimes, explicitly labeled as engineering approximations rather than theorems.
2. **Empirical-vs-proxy gap analysis** showing that, where analytic proxies overestimate empirical refreshing time, the gap ranges from roughly $8\times$ (Markov) to $125\times$ (long-memory); in the IID and burst regimes the empirical values instead exceed the analytic proxy, reflecting finite-sample autocorrelation fluctuations and, for burst, forward-window counting effects at the chosen window length. These directional gaps indicate where tighter family-specific characterisations are most needed.
3. **A (τ, r) sensitivity landscape** that maps the quantum accuracy proxy as a function of correlation parameters, identifying regions where the proxy accuracy is highest; this landscape is the paper's central diagnostic and hypothesis-generation artifact.
4. **Regime-dependent robustness hypotheses:** the proxy model predicts a hypothesized quantum-favourable region under mild to moderate correlation that narrows predictably as correlation strengthens. These are proxy-level hypotheses intended for future algorithm-level validation, not demonstrated quantum performance.

Sections 2–3 review related work and introduce the proxy model; Section 4 specifies the correlation proxies; Section 5 describes the experimental setup; Section 6 presents nine experiments (seven on synthetic streams, a two-dataset real-stream sanity check, and an ablation suite); Section 7 discusses implications, the analytic–empirical divergence, and the scope of the diagnostic; Section 8 systematises threats to validity; Section 9 outlines future work; Section 10 concludes.

2. Related Work

2.1. Classical Streaming Algorithms

Classical data streaming has been studied extensively since the seminal work of Alon et al. [3] on frequency moments [37]. Key developments relevant to this paper include the Count-Min Sketch of Cormode and Muthukrishnan [16] for frequency estimation, the Frequent Directions algorithm of Ghashami et al. [21] for streaming low-rank approximation, and the sketching-as-linear-algebra framework of Woodruff [49]. On the learning side, Shalev-Shwartz [44] provides the standard reference for online convex optimisation; Polyak and Juditsky [39] introduced Polyak–Ruppert averaging, which we use as a second classical baseline; and Weinberger et al. [47] established the feature-hashing technique underlying our bounded-memory classifier.

The behaviour of streaming algorithms under structured temporal correlation has received comparatively less attention. Concept drift is surveyed by Gama et al. [20] and Ditzler et al. [18]; ADWIN [11] is the canonical adaptive-window solution. Broader online-learning treatments appear in MOA [12], Hoi et al. [27], and Gama [19]; the most recent systematic drift review is Arora et al. [4]; adaptive random forests [24] serve as the drift-aware classical reference. Recent work on non-stationary streaming includes elastic online deep-learning pipelines [45], bi-dynamic-distribution learning [51], and evolving-prototype classifiers for imbalanced streams [50]. Non-stationary online optimisation is analysed in Besbes et al. [9] and Ben-David et al. [8]; Count-Min refinements appear in Ben Mazziane et al. [6] and Cormode et al. [17]; foundational self-similarity measurements are in Park and Willinger [38].

2.2. Quantum Advantages for Data Processing

Quantum algorithms for data processing include quantum SVMs [41], quantum PCA [35], the quantum singular value transformation [22] with block-encoding subroutines [14]. Critical perspectives on access-model dependencies appear in Aaronson [1] and Tang [46]; qRAM feasibility is discussed in Giovannetti et al. [23]; surveys include Chen et al. [15] and Schuld and Petruccione [42]. Quantum advantages depend critically on data structure and access [29]; an experimental demonstration of exponential advantage appears in Huang et al. [30]. On quantum streaming: Kallaugher [31] established the first exponential quantum-classical space separation for a natural streaming problem; Kallaugher et al. [32] provided a black-box quantum-sketch construction for classical algorithm designers. The quantum oracle sketching framework of Zhao et al. [52] avoids qRAM entirely and supports exponential separations for classification and dimension-reduction tasks.

2.3. Non-IID Data in Quantum Computing

Arunachalam and de Wolf [5] study quantum sample complexity under the IID PAC assumption. The quantum oracle sketching framework [52] is, to our knowledge, the first to address noisy, correlated, and time-varying classical data in a quantum streaming setting. We introduce operational proxies inspired by their correlated-data discussion across five structured regimes; the Markov mixing time follows product-chain results [34] and the long-memory statistics follow fGn theory [7,36].

3. Preliminaries

3.1. Quantum Oracle Sketching

The originating work [52] shows that a polylogarithmic-size quantum computer ($\mathcal{O}(n)$ qubits for n-dimensional data, $N = 2^n$) can perform classification and dimension reduction on massive classical data by processing random samples on the fly via quantum oracle sketching; Boolean function learning is one concrete instantiation. For several concrete tasks, the classical memory requirement scales polynomially in N, yielding an exponential memory gap. Their correlated-data extension uses a structured data-generation model that accounts for temporal dependence; we adopt the terms *refreshing time* (decorrelation scale) and *repetition number* (expected duplication) as our operational labels for the key correlation parameters, noting that our definitions (Section 3.2) are inspired by but not necessarily identical to the exact quantities in their framework. Classical hardness results are task-specific, not a universal lower bound.

3.2. Heuristic Effective-Sample Model

Proxy status. The quantities τ and r defined below are *paper-defined operational proxies* motivated by, but not formally equivalent to, the correlation parameters of the exact framework; Section 3.3 separates precisely what this paper inherits from established theory, what it assumes, and what it constructs heuristically. We introduce a simplified effective-sample proxy that adapts the qualitative structure of the framework's correlated-data analysis into a tractable form for computational exploration across diverse correlation families. The choice $C{=}2$, $\delta{=}0.05$ in Eq. (2) rescales accuracy contours uniformly without deforming the qualitative regime ordering: changing C or δ shifts every contour along the hyperbolic family $\tau r \propto (C^2 \log(1/\delta))^{-1}$, preserving the relative ranking of all regimes—although the *absolute* location of any proxy–baseline crossover does shift with these constants, as quantified in Section 6.9 (Table 7). Practitioners interpreting the landscape quantitatively should treat absolute contour values as indicative rather than universal.

Definition 1 (Refreshing Time – operational proxy)**.** *We define the* operational refreshing time τ *of a data stream as the smallest lag at which the normalized autocorrelation drops below* $1/e$*. This is an empirically measurable quantity that serves as a proxy for the decorrelation rate of the stream.*

Definition 2 (Repetition Number – operational proxy)**.** *We define the* operational repetition number r *as* 1 *plus the expected number of* exact duplicate *copies of each sample within a forward window of length* W*. Throughout this paper we use* $W = 20$ *in the empirical estimator; this is the definition that Fig. 4(b) reports and matches the analytic family-level* r *in units.*

We model the effective independent sample count as:

$$T_{\text{eff}} = T/(\tau \cdot r), \tag{1}$$

and use the following heuristic oracle error proxy:

$$\epsilon \le C\sqrt{\frac{n\log(1/\delta)}{T_{\text{eff}}}}, \tag{2}$$

where $C = 2$ and $\delta = 0.05$ throughout our experiments. Under IID data, $\tau = 1$ and $r = 1$, recovering the standard scaling $\epsilon \sim \sqrt{n/T}$.

The heuristic classification accuracy proxy used in our experiments is:

$$\text{acc\_proxy} = \max\Big(0.5,\ 1 - \epsilon^2\Big), \tag{3}$$

where the floor of 0.5 represents the random-guess baseline for binary classification. When $\epsilon^2 > 0.5$ (as occurs in the long-memory regime), the proxy saturates at 0.5, reflecting that the model predicts no useful signal. This is an engineering surrogate that translates oracle approximation quality into a classification metric for visualization.

3.3. Relationship Between Operational Proxies and Quantum Oracle-Sketching Theory

This subsection separates what the present study inherits from established theory, what it assumes, and what it constructs heuristically; Table 1 summarises the provenance and epistemic status of every quantity used in the paper.

Established theory

The exact framework [52] proves three results that this paper takes as given: (i) a polylogarithmic-size quantum computer ($\mathcal{O}(n)$ qubits, $n = \log_2 N$) suffices to construct oracle sketches of massive classical data streams; (ii) under the framework's hierarchical model of correlated data generation, the oracle-construction sample complexity is multiplied by the repetition number of the process—the refreshing time enters only by delimiting the window over which repetition is counted, and carries no independent penalty; and (iii) for specific tasks, classical machines of size $\mathcal{O}(N^c)$, for any constant $c < 1$, cannot match the quantum performance [3,31]. All three are theorems about the framework's constructions and hard instances; none has been verified on physical hardware, and this paper adds no new theory.

Assumptions

We assume (a) that a stream's temporal dependence can be usefully summarised by two scalars—a decorrelation scale and a duplication rate; (b) that binarised feature streams (Section 5) retain enough of the correlation structure of the underlying process for such scalars to be meaningful; and (c) that a task-level statistical generalisation rate evaluated at an effective sample count is a useful stand-in for end-to-end pipeline behaviour. None of these assumptions is inherited from the exact framework; they are modelling choices of this paper.

Heuristic constructions introduced in this paper

Two deliberate simplifications distinguish Eqs. (1)–(2) from the exact guarantees. First, in the exact framework the correlation overhead enters through the repetition number alone, whereas Eq. (1) additionally divides by τ, in the manner of a classical effective-sample-size correction ($T_{\text{eff}} \approx T/\tau_{\text{int}}$). Our T_{eff} is therefore deliberately *more conservative* than the exact overhead and should be read as a pessimistic effective-information budget, not as a prediction of the framework's theorems. Second, the $\sqrt{n/T_{\text{eff}}}$ template in Eq. (2) is the statistical generalisation rate for the n-dimensional linear classification task studied here (VC dimension $n+1$) applied to the effective sample count; it is *not* the sample cost of the exact oracle-construction procedure, which for a uniform marginal over $N = 2^n$ inputs scales linearly in N. Throughout, *oracle error* accordingly denotes the approximation error of our heuristic model's decision rule, not the construction error of the exact sketching procedure; the two views coincide numerically at our operating point ($n = 10$, $T = 10^4$) but scale differently in n (see Experiment 6).

Memory references

For quantum memory, the exact framework establishes that $\mathcal{O}(n)$ qubits suffice for oracle construction. On the classical side, it establishes task-specific hardness results in which classical machines of size $\mathcal{O}(N^c)$, for any constant $c < 1$, cannot match the quantum performance. Throughout this paper we use $\Omega(\sqrt{N})$ as a *conservative representative* of this polynomial-in-N family, in the spirit of classical streaming space bounds [3,31]; it is a theoretical reference only. It is *not* an empirically witnessed floor for any learner studied here: no experiment in this paper measures a memory lower bound for online SGD, averaged SGD, or Count-Min, and for the 10-dimensional linear classification task used in our experiments the classical learners are sample-limited rather than memory-limited once M exceeds a small threshold (Section 6.2). The $\Omega(\sqrt{N})$ curve in our figures therefore illustrates the worst-case asymptotic separation established by theory, not an observed empirical gap.

Scope limitations

The operational τ and r are measurement conventions, not theoretical quantities: they are computable from any stream but are not formally equivalent to—and, as Section 7.3 shows, can diverge substantially from—the refreshing time and repetition number of the

Table 1. Provenance and epistemic status of the quantities used in this paper. "Established" items are theorems of the originating framework or standard learning theory; "introduced here" items are operational constructions of this paper; the final column states what is *not* theoretically guaranteed.

Quantity	Status	Not guaranteed
$\times R$ sample overhead; $\mathcal{O}(n)$ qubits; $\mathcal{O}(N^c)$ classical hardness	established theory	hardware validation; applicability beyond the framework's hard instances
τ ($1/e$ ACF crossing)	introduced here (operational)	equivalence to the framework's refreshing time
r (W-window duplicate count)	introduced here (operational)	equivalence to the framework's repetition number
$T_{\text{eff}} = T/(\tau r)$	introduced here (conservative)	correspondence to any exact sample-complexity theorem
$\epsilon \le C\sqrt{n \log(1/\delta)/T_{\text{eff}}}$	standard VC-type task rate, repurposed here	equality with oracle-construction cost (scales differently in n)
acc_proxy $= \max(0.5, 1-\epsilon^2)$	introduced here (visualisation surrogate)	correspondence to any implemented algorithm's accuracy
$\Omega(\sqrt{N})$ memory curve	conservative theoretical reference	empirical memory floor for SGD, averaged SGD, or Count-Min

exact framework, whose definitions are tied to a hierarchical generative model that real streams need not follow. Consequently, positions on the (τ, r) landscape are diagnostic coordinates, and every "favourable"/"unfavourable" designation in this paper is a statement about the proxy model evaluated at those coordinates, not about any realised quantum computation. Establishing formal equivalence, or bounding the divergence, is follow-on theoretical work (Section 9).

4. Correlation Proxies for Specific Data Families

We specify heuristic expressions for the operational τ and r under five concrete correlation regimes. Each is an engineering proxy motivated by the autocorrelation or mixing structure of the regime; tighter proofs of equivalence to the exact framework's correlation parameters are follow-on work.

4.1. Regime 1: IID Baseline

Under IID sampling from a uniform distribution over $\{0,1\}^n$:

$$\tau_{\text{IID}} = 1, \qquad r_{\text{IID}} = 1, \qquad T_{\text{eff}} = T. \tag{4}$$

This serves as the reference against which all other regimes are compared.

4.2. Regime 2: Markov Switching

Consider a Markov chain on $\{0,1\}^n$ where each bit flips independently with probability $(1-\rho)/2$ at each step, for correlation parameter $\rho \in [0,1)$. The stationary distribution is uniform.

The mixing time of this product chain is $\Theta(n \log n/(1-\rho))$. We use the simplified engineering proxy:

$$\tau_{\text{Markov}}(\rho, n) \approx \frac{n}{1-\rho}, \tag{5}$$

which preserves the dominant divergence as $\rho \to 1$ while suppressing logarithmic and model-specific constants. *Provenance:* the $1/(1-\rho)$ scaling is theory-motivated (mixing-time asymptotics); the omission of the $\log n$ factor is an engineering simplification; the equivalence to the exact framework's correlation parameters is unvalidated. Since samples from a Markov chain are not repeated exactly, $r_{\text{Markov}} = 1$, giving $T_{\text{eff}} \approx T(1-\rho)/n$.

4.3. Regime 3: Seasonal Drift

Consider a stream where bit i at time t is drawn as Bernoulli with probability $p_i(t) = \frac{1}{2} + \alpha \sin(2\pi t/P + \phi_i)$, where P is the period, $\alpha \in (0, 0.5)$ is the drift strength, and ϕ_i are fixed phase offsets.

The seasonal regime is *distributional drift*: samples at distinct times are conditionally independent given t, making this non-IID only in the weak sense of non-stationary marginals. It is included because both the quantum oracle construction and classical learners are affected by marginal drift, and because it is a common practical stream pattern.

We propose the decorrelation proxy:

$$\tau_{\text{Seasonal}}(P, \alpha) = \max(1,\ P \cdot \alpha), \tag{6}$$

lower-bounded by 1 to ensure $\tau \geq \tau_{\text{IID}}$ in all cases. *Provenance:* the product $P\alpha$ captures the intuition that decorrelation scales with both period and drift strength; the functional form $\max(1, P\alpha)$ is an engineering convenience with no formal derivation. Since the distribution changes continuously, $r_{\text{Seasonal}} = 1$.

4.4. Regime 4: Burst Repetition

In the burst model, at each step, with probability p_b the previous sample is exactly repeated for an additional L consecutive copies (i.e., the burst contributes L duplicate observations). The expected number of observations per effective fresh sample is therefore:

$$r_{\text{Burst}}(L, p_b) = 1 + p_b \cdot L. \tag{7}$$

Provenance: the formula follows directly from the model definition (theory-motivated); the assumption that duplicates contribute no new information is exact for our synthetic generator but would require validation on real burst-prone streams. Since bursts do not affect the mixing of fresh samples, $\tau_{\text{Burst}} = 1$, giving $T_{\text{eff}} = T/(1 + p_b L)$.

4.5. Regime 5: Long-Range Dependence

For streams driven by fractional Gaussian noise (fGn) with Hurst parameter $H \in (0.5, 1)$, the autocorrelation of the *underlying Gaussian process* decays as a power law, and integrated autocorrelation grows as T^{2H-1}. We adopt as our proxy the horizon-dependent correlation penalty:

$$\tau_{\text{LRD}}(H, T) = T^{2H-1}, \tag{8}$$

yielding $T_{\text{eff}} = T^{2(1-H)}$.

Proxy status

Equation (8) is an *upper-bound proxy*. In our construction the stream is a *binarized* fGn (each bit is a thresholded fGn path); binarization converts Gaussian autocorrelation $\gamma(k)$ into binary autocorrelation $(2/\pi)\arcsin(\gamma(k))$, which is strictly smaller in absolute value. The effective decorrelation of the binary stream is therefore *faster* than Eq. (8) suggests, so the proxy is conservative. Deriving a tighter proxy that accounts for the arcsine transformation is deferred to future work. Unlike the other regimes, this proxy depends on the stream length T, reflecting the property that the underlying driving process has no finite decorrelation time.

4.6. Summary

Table 2 summarizes the proxy parameters and heuristic MSE proxy values for each regime, computed from the single master formula $\epsilon^2 = \min(1,\ C^2 n \log(1/\delta)/T_{\text{eff}})$ with $C = 2$ and $\delta = 0.05$.

Table 2. Summary of heuristic correlation proxies and MSE proxy values (ϵ^2) for each regime, with $T = 10{,}000$, $n = 10$, $C = 2$, and $\delta = 0.05$. The factor $\kappa = C^2 \log(1/\delta) \approx 12$.

Regime	τ	r	T_{eff}	MSE proxy
IID	1	1	10,000	0.012
Markov (ρ=0.7)	$n/(1-\rho) \approx 33$	1	≈ 300	0.40
Seasonal (P=100, α=0.3)	$\max(1, P\alpha) = 30$	1	≈ 333	0.36
Burst (L=10, p=0.3)	1	$1+pL = 4$	2,500	0.048
Long-memory (H=0.8)	$T^{2H-1} \approx 251$	1	≈ 40	1.0 (saturated)

5. Experimental Framework

Throughout this section, the comparison is between (1) empirical bounded-memory classical learners evaluated on a simple linear classification task, and (2) a theory-inspired heuristic proxy for quantum oracle sketching performance; the empirical classical baselines are not the hard problem instances of the exact framework's lower-bound constructions.

5.1. Data Generation and Diagnostics

Data streams over $\{0,1\}^n$ with $n = 10$ ($N = 1{,}024$) and $T \leq 20{,}000$. Target task: binary classification with random linear separator $f(x) = \mathbf{1}[\langle x - \frac{1}{2}\mathbf{1}, w\rangle > 0]$, $w \in \mathbb{R}^n$ fixed. Parameter values follow Section 4: IID (uniform), Markov ($\rho = 0.7$), Seasonal ($P = 100$, $\alpha = 0.3$), Burst ($L = 10$, $p_b = 0.3$), Long-memory ($H = 0.8$).

Empirical diagnostics: τ is the smallest lag where normalized autocorrelation (averaged over bits) drops below $1/e$; r is 1 plus the mean exact-duplicate count in a $W = 20$ forward window.

5.2. Classical Baselines

Three memory-bounded streaming classifiers cover the key axes:

- **Online SGD**: logistic regression with feature hashing [44,47]; memory $\mathcal{O}(M)$.
- **Averaged SGD**: Polyak–Ruppert running average $\bar{w}_t = \frac{1}{t}\sum_s w_s$ [39]; lower variance than plain SGD.
- **Count-Min hash classifier**: Count-Min Sketch [16] as a label-averaging lookup table; serves as a collision-limited non-gradient reference (near-chance for $M \ll N$).

Memory budgets range from $M = 8$ to $M = 256$.

5.3. Quantum Performance Proxies

Rather than simulating quantum circuits (which would require exponential classical resources at meaningful scale), we compute proxy quantities from the effective-sample model of Section 3.2:

- Oracle MSE proxy: $\epsilon^2 = C^2 n \log(1/\delta)/T_{\text{eff}}$, with $C = 2$, $\delta = 0.05$.
- Classification accuracy proxy: acc_proxy $= \max(0.5,\ 1 - \epsilon^2)$ (Eq. 3).
- Quantum memory: $\mathcal{O}(n)$ qubits [52].

Justification of proxy constants and forms

Three design choices in the proxy expressions warrant explicit discussion. (i) The constant $C = 2$ in $\epsilon^2 = C^2 n \log(1/\delta)/T_{\text{eff}}$ is a conservative scaling factor consistent with concentration-inequality constants in the quantum oracle sketching literature; alternative values $C \in \{1, 1.5, 2.5\}$ scale the proxy curve uniformly and preserve all relative orderings between regimes, though the absolute proxy–baseline crossover location shifts with C (Table 7). (ii) The confidence parameter $\delta = 0.05$ matches the 95% reporting convention used throughout the paper; a $10\times$ change in δ shifts $\log(1/\delta)$ by a factor of only $\approx$1.5, which is small relative to the cross-regime differences in T_{eff} but visible in the absolute crossover location (Table 7). (iii) The seasonal proxy $\tau = \max(1, P\alpha)$ is an engineering convenience

Table 3. Reproducibility summary: fixed experimental parameters.

Parameter	Value / method
Seeds per experiment	10 (indices 0–9)
Forward window W	20
Proxy constants	$C = 2$, $\delta = 0.05$, $\kappa = C^2 \log(1/\delta) \approx 12$
τ_{emp} estimator	smallest lag k with mean bitwise ACF $< 1/e$
r_{emp} estimator	1+ mean exact-duplicate count in W-forward window
Wilcoxon pairing	per-seed classical accuracy vs. same-seed proxy value
Stochastic quantities	classical accuracy, τ_{emp}, r_{emp} (10-seed, 95% CI)
Deterministic quantities	T_{eff}, ϵ^2, acc_proxy, (τ, r) landscape

encoding that periodic streams effectively reset every $\sim P\alpha$ samples; the long-memory proxy uses an arcsine correction following Beran [7] to account for binarisation, and is acknowledged as conservative. Section 6.9 reports an explicit sensitivity analysis under four alternative τ estimators and three target functions, and shows that the qualitative regime ordering is preserved.

Measured vs. proxy

Empirically measured (10 seeds, 95% CI): classical accuracy, τ_{emp}, r_{emp}. **Proxy-estimated** (deterministic): $T_{\text{eff}} = T/(\tau r)$, $\epsilon^2 = \min(1, \kappa n/T_{\text{eff}})$ with $\kappa \approx 12$, acc_proxy $= \max(0.5, 1 - \epsilon^2)$. All figures clearly distinguish the two categories; deterministic landscapes (Fig. 3) have no seed dependency.

Reproducibility parameters

Table 3 consolidates all fixed experimental constants for independent reimplementation.

6. Results and Discussion

We now present nine experiments. Experiments 1–7 are visualisation experiments on the synthetic non-IID regimes, each accompanied by multi-seed classical baselines on the same streams: Experiments 1–2 compare accuracy trajectories; Experiment 3 presents the analytic (τ, r) sensitivity landscape; Experiment 4 contrasts empirical and analytic correlation parameters; Experiment 5 sweeps Markov correlation strength; Experiment 6 examines dimension scaling; Experiment 7 shows rolling-accuracy dynamics over time. Experiment 8 is a real-stream sanity check on two NAB datasets (Section 6.8); Experiment 9 is a sensitivity-ablation suite (Section 6.9).

6.1. Experiment 1: Classification Accuracy vs. Sample Count

Figure 1 shows that all classical baselines converge to 0.88–0.92 accuracy with narrow CI bands. Averaged SGD (Polyak–Ruppert, 39) produces smoother trajectories, particularly in the burst regime; Count-Min saturates near 0.55–0.60 due to hash collisions at $M \ll N$. The MSE proxy (Panel b) decays as $1/T_{\text{eff}}$: IID and burst show fast decay; for Markov, seasonal, and long-memory, analytic τ inflates ϵ^2 enough that the accuracy proxy falls below classical (Section 6.4 quantifies this).

6.2. Experiment 2: Classical Accuracy vs. Memory Budget

Figure 2 shows that classical accuracy saturates at $M \geq 16$, confirming that the SGD learner is sample-limited rather than memory-limited at this scale. Under the IID modelling assumptions, the IID proxy curve ($\sim$0.99) lies above the empirical classical curves on this task—a proxy-side ordering, not an implemented algorithmic comparison; the $\Omega(\sqrt{N})$ memory reference is a conservative theoretical stand-in for the task-specific polynomial-in-N classical bounds (Section 3.3).

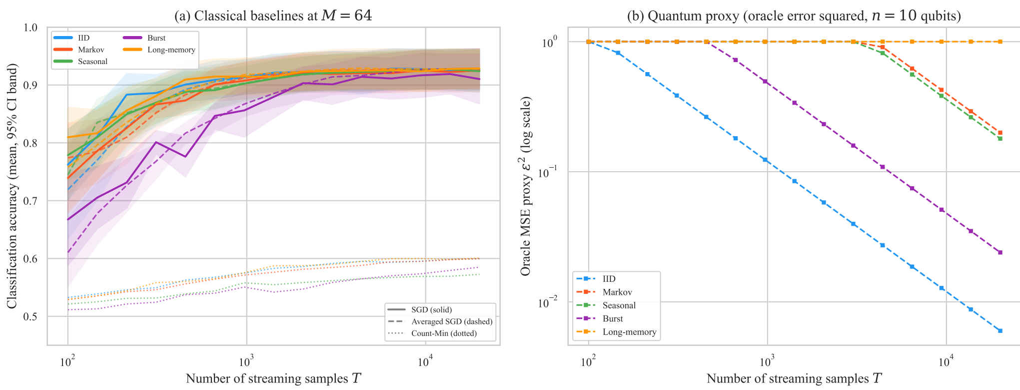

Figure 1. Classification accuracy and quantum proxy error vs. stream length T (10-seed mean, 95% CI bands). **(a)** Three classical baselines at $M = 64$: online SGD (solid), averaged SGD (dashed), Count-Min (dotted). **(b)** Quantum oracle MSE proxy ϵ^2 ($n = 10$ qubits), log-log scale.

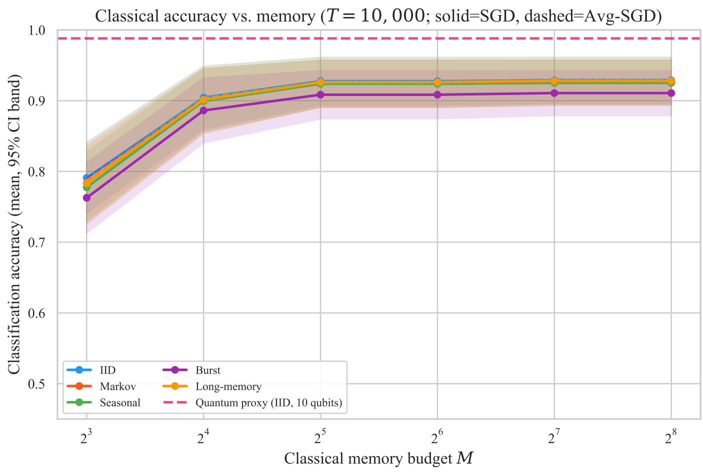

Figure 2. Classical online SGD accuracy vs. memory M ($T = 10{,}000$) compared with the quantum IID proxy (dashed). Classical accuracy saturates at $M \geq 16$; under the IID modelling assumptions, the IID proxy lies above the empirical classical curves on this task.

6.3. Experiment 3: Sensitivity Landscape in (τ, r) Space

T_{eff} decreases diagonally as τ and r increase; accuracy contours follow hyperbolas in $\tau \cdot r$ space. IID and burst fall in the high-accuracy corner; Markov and seasonal sit near the 0.7–0.8 contour; long-memory falls in the low-accuracy zone. Researchers can place estimated (τ, r) values on the landscape to identify comparable synthetic regimes for follow-on empirical or algorithm-level validation.

6.4. Experiment 4: Empirical vs. Proxy Correlation Parameters

Three findings emerge from Fig. 4. First, analytic τ systematically overshoots empirical τ: Markov (33 vs. 4, $\approx 8\times$), long-memory (251 vs. 2, $\approx 125\times$, explained by the arcsine relationship from binarisation [7,36]), and seasonal (30 vs. 2); IID and burst show the opposite sign from finite-sample fluctuations. Second, unit-aligned r values agree well: burst empirical $r \approx 5.6$ vs. analytic $r = 4$ (factor $1.4\times$, attributable to window-length choice); all other regimes near 1. Third, IID ($\tau_{\text{emp}} \approx 2$ vs. $\tau_{\text{analytic}} = 1$) sets the finite-sample noise floor. These findings confirm that (τ, r) are qualitatively informative but quantitatively loose; family-specific refinements would enlarge the predicted favourable region, and Section 7.3 explains the mechanisms behind the directional gaps.

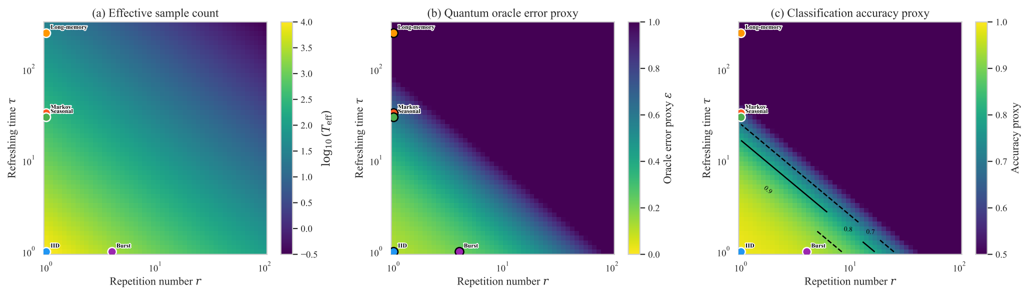

Figure 3. Sensitivity landscape in (τ, r) space ($T = 10{,}000$, $n = 10$). **(a)** $T_{\text{eff}} = T/(\tau r)$, $\log_{10}$ scale. **(b)** Oracle error proxy ϵ. **(c)** Classification accuracy proxy with contours at 0.7, 0.8, 0.9; colored dots mark each regime's analytic (τ, r). The landscape is a hypothesis-generation map: contour values are proxy estimates under the modelling assumptions of Section 3.2, not measured quantum performance.

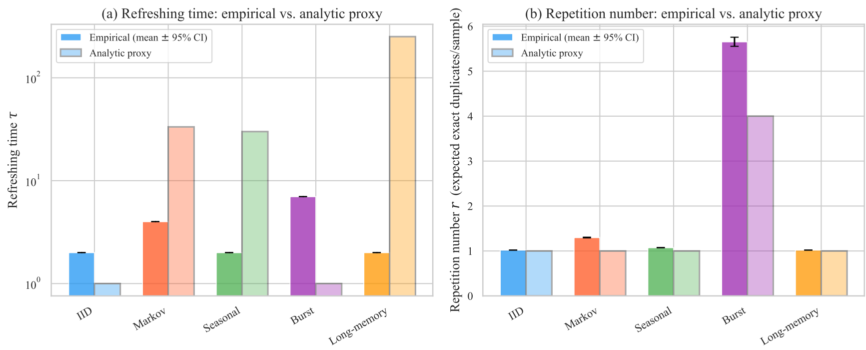

Figure 4. Empirical vs. analytic proxy τ and r (10-seed mean $\pm$ 95% CI). **(a)** Refreshing time τ: analytic exceeds empirical for Markov, seasonal, long-memory; opposite for IID and burst. **(b)** Repetition number r: both quantities in units of expected exact duplicate copies per sample within a $W = 20$ forward window.

6.5. Experiment 5: Correlation Strength Sweep

Figure 5 examines how the classical-vs-proxy separation evolves as the Markov correlation strength ρ is varied continuously from 0 (IID) to 0.95 (strong correlation), comparing the classical SGD baseline against the quantum proxy using both the *analytic* τ and the *empirically measured* τ.

Panel (a) reveals a key insight: using the analytic proxy τ, the quantum performance proxy crosses below classical performance at $\rho \approx 0.3$, suggesting the proxy-model crossover occurs under moderate Markov correlation. However, using empirical τ instead, the empirical-τ proxy remains above 0.9 for $\rho \leq 0.8$ and above the classical baseline across most of the sweep. Under the empirical-proxy model, the paired proxy $-$ classical difference shrinks monotonically from $+0.042$ at $\rho = 0$ (Hodges–Lehmann estimate; exact 95% CI $[+0.011, +0.087]$) to $+0.019$ at $\rho = 0.68$ (CI $[-0.011, +0.069]$, spanning zero) and reverses sign to -0.046 by $\rho = 0.88$ (CI $[-0.079, -0.003]$). The mean proxy and classical curves cross at $\rho^* \approx 0.82$ under the paper's operating constants (interpolated on the sweep grid; Section 6.9 quantifies how ρ^* moves with C and δ), and the hypothesized proxy-favourable region is closed by $\rho \approx 0.88$, where even the empirical autocorrelation becomes severe. Table 4 reports the full interval estimates and effect sizes.

Panel (b) quantifies the source of this discrepancy: the analytic proxy $\tau = n/(1-\rho)$ grows as a pole at $\rho = 1$, while the empirical τ grows much more gradually. The shaded proxy–practice gap widens by over an order of magnitude at $\rho = 0.9$. This gap is the primary driver of the pessimistic quantum performance proxies observed in Experiments 1–3, and its characterization is one of the central contributions of this paper.

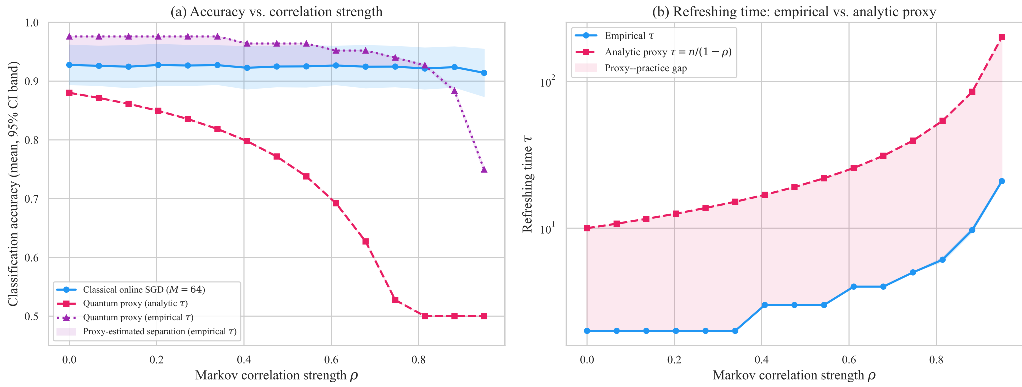

Figure 5. Markov correlation strength sweep (10-seed mean with 95% CI bands). **(a)** Classification accuracy: classical online SGD at $M = 64$ (blue) is approximately flat at $\sim$0.92 across the sweep. The quantum accuracy proxy using the *analytic* $\tau = n/(1-\rho)$ (pink, dashed) degrades and crosses below classical at moderate ρ near the lower end of the sweep. The quantum proxy using the *empirical* τ (purple, dotted) remains above classical until the mean curves cross at $\rho^* \approx 0.82$, with the sign reversal decisive by $\rho \approx 0.88$ (Table 4), indicating that tighter family-specific characterisations would enlarge the region in which the proxy model predicts a favourable separation. **(b)** Refreshing time: the proxy–practice gap (shaded) grows with ρ, showing that the analytic proxy is increasingly conservative under stronger correlation. CI bands are narrow throughout, confirming that the main trends are seed-stable.

What these tests do and do not establish

These comparisons pair an empirical, stochastic classical learner with a model prediction, not with a second implemented algorithm: for each ρ, the paired observations are the per-seed classical accuracy and the same seed's proxy value, which is a closed-form function of that seed's empirically measured τ. The proxy's seed-to-seed variability is therefore inherited entirely from the τ estimator (negligible at $\rho \leq 0.68$, ± 0.004 at $\rho = 0.88$), not from any stochastic training process. The tests consequently quantify whether the sign of the classical $-$ proxy offset is stable across seeds—they do not, and cannot, establish that one algorithm outperforms another, and no quantum algorithm is implemented anywhere in this comparison. We accordingly interpret Table 4 through the effect sizes: the paired difference is large and positive at mild correlation ($r_{\mathrm{rb}} \geq +0.71$, exact CIs excluding zero for $\rho \leq 0.47$), indistinguishable from zero at $\rho = 0.68$, and large and negative at $\rho = 0.88$ ($r_{\mathrm{rb}} = -0.75$, CI excluding zero). The unadjusted p-values are reported for completeness; under a Holm correction across the five sweep points none falls below the 0.05 family-wise threshold at $n = 10$ seeds, so we treat the monotone sign reversal and its interval estimates—not statistical significance—as the finding.

6.6. Experiment 6: Dimension Scaling

Figure 6 illustrates the exponential scaling of the *theoretical* quantum-vs-classical memory references with data dimension. No memory lower bound is measured empirically in this experiment: the curves in panel (a) plot cited theoretical quantities, and no result here constitutes an empirical memory floor for online SGD, averaged SGD, or Count-Min.

Panel (a) shows that at $n = 14$ ($N = 16{,}384$) the representative classical memory reference exceeds 128 while the quantum reference is only 14 qubits — a separation that is exponential in n at the level of the cited bounds (the $\Omega(\sqrt{N})$ floor is a theoretical worst-case reference, not an empirical floor measured for any baseline here; see Section 3.3). Panel (b) shows that classical SGD accuracy is approximately stable across $n \in [6, 14]$, while the quantum IID proxy is consistently near 1.0. This flatness is a property of the task-level $\sqrt{n/T_{\mathrm{eff}}}$ rate; under the exact framework's oracle-construction cost, which grows linearly in $N = 2^n$, the fixed horizon $T = 10^4$ is already comparable to N at $n = 14$, so the flat

Table 4. Classical-vs-proxy comparison across representative Markov correlation strengths ρ: empirical classical online-SGD accuracy at $M = 64$ and the heuristic quantum accuracy proxy using the empirically measured τ (10-seed mean $\pm$ 95% CI half-width). $\hat{\Delta}$ is the Hodges–Lehmann estimate of the paired (proxy $-$ classical) difference with its exact signed-rank 95% CI; r_{rb} is the matched-pairs rank-biserial correlation; p is the two-sided paired Wilcoxon signed-rank p-value, reported descriptively. A positive $\hat{\Delta}$ means only that the proxy estimate exceeds the empirical classical baseline, not realised quantum algorithmic superiority. Under a Holm correction across the five sweep points no comparison falls below the 0.05 family-wise level at $n = 10$ seeds (smallest adjusted $p = 0.098$); the evidence for the crossover is the monotone sign reversal of $\hat{\Delta}$ and its interval estimates, not any single test.

ρ	Classical SGD	Proxy (emp. τ)	$\hat{\Delta}$ [95% CI]	r_{rb}	p
0.00	0.928 ± 0.034	0.976 ± 0.000	$+0.042$ $[+0.011, +0.087]$	$+0.82$	0.020
0.27	0.927 ± 0.034	0.976 ± 0.000	$+0.042$ $[+0.012, +0.090]$	$+0.78$	0.027
0.47	0.925 ± 0.034	0.964 ± 0.000	$+0.031$ $[+0.002, +0.080]$	$+0.71$	0.049
0.68	0.925 ± 0.036	0.952 ± 0.000	$+0.019$ $[-0.011, +0.069]$	$+0.35$	0.375
0.88	0.924 ± 0.034	0.884 ± 0.004	-0.046 $[-0.079, -0.003]$	-0.75	0.037

trend should not be read as construction feasibility at higher dimensions (Section 3.3). The quantum Markov proxy *decreases* with dimension because $\tau_{\mathrm{Markov}} = n/(1-\rho)$ grows linearly with n; family-specific proxy refinements are therefore most useful at higher dimensions.

6.7. Experiment 7: Rolling Accuracy Dynamics

Figure 7 confirms the regime-dependent pattern of Experiments 1 and 5: IID and burst show higher proxy-estimated trajectories; Markov and seasonal show a visible gap attributable to the conservative analytic τ (a tighter family-specific proxy would shift the trajectory upward, as the Markov sweep confirms); long-memory exhibits the 0.5 saturation floor and persistent low-frequency classical oscillations characteristic of long-range dependence.

6.8. Experiment 8: Real-Stream Sanity Check

The synthetic experiments above leave open whether the empirical τ and r estimators apply usefully to real streams. To answer this we apply the proxy framework end-to-end to two publicly available real streams from the Numenta Anomaly Benchmark [2] (NAB; MIT licence) and report what is observed. No claim of quantum advantage is made; the contribution of this experiment is the successful application of the diagnostic workflow outside the synthetic regime, not a comparison against a realised quantum algorithm. The two streams were chosen *before* examining the results to span very different correlation regimes (transport-traffic seasonality versus industrial-sensor drift), and we pre-committed to reporting whichever placement and baseline outcomes emerged.

Datasets

(i) *NYC Taxi*: $T = 10{,}320$ half-hour pickup counts in New York City (2014-07-01 to 2015-01-31). Combines strong daily/weekly seasonality with documented burst events (NYC Marathon, Thanksgiving, snowstorms); overlaps the synthetic seasonal + burst + low-order Markov regimes. (ii) *Machine Temperature*: $T = 22{,}695$ five-minute readings from an industrial machine sensor preceding a documented system failure. Exhibits slow drift over hours-to-days plus localised spike events; overlaps the long-memory + burst regimes.

Binarisation choice

Each continuous series is encoded into $n = 6$ binary features (matched to Section 6.9): a 3-bit time-of-day bucket, a weekend flag, and two trend bits (current and lag-1 indicator of whether the value exceeds a 7-day rolling median; the rolling window is 336 bins for

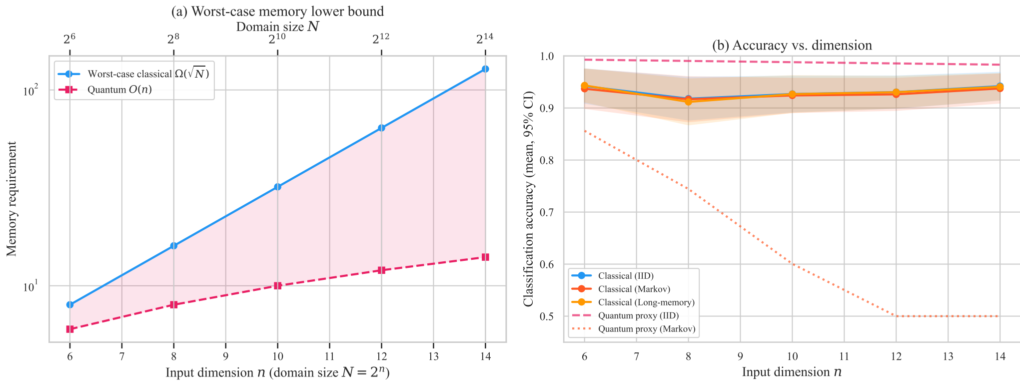

Figure 6. Dimension scaling analysis (10-seed mean, 95% CI bands). **(a)** Theoretical worst-case memory reference: representative classical lower bound $\Omega(\sqrt{N}) = \Omega(2^{n/2})$ from the cited task-specific hardness results vs. quantum $\mathcal{O}(n)$. The separation is exponential in n at the level of theory; it is not a witnessed empirical floor for any classical baseline studied here (Section 3.3). **(b)** Classification accuracy vs. dimension n: classical online-SGD accuracy for IID, Markov, and long-memory data (solid lines) compared with quantum accuracy proxies for IID and Markov (dashed). The quantum IID proxy is consistently near 1.0 (a task-level generalisation trend, not oracle-construction feasibility; Section 3.2); the quantum Markov proxy degrades with dimension (because $\tau_{\mathrm{Markov}} = n/(1-\rho)$ grows with n, reducing T_{eff}), while classical accuracy is approximately stable across the range studied.

NYC Taxi at 30-min resolution and 2,016 bins for Machine Temperature at 5-min resolution). The target y_t is a simple rare-event classification proxy: $y_t = 1$ iff the next-bin value is at or above the 80th percentile of the full series. *This is a per-stream high-activity threshold and not the original NAB anomaly label.* The 80th percentile is chosen because it produces a non-trivial ∼20%/80% class imbalance that exposes both majority-baseline behaviour and the rare-event regime; this target is intentionally imbalanced and not a balanced classification task. To prevent accuracy from masking majority-class memorisation, we report balanced accuracy and F_1 of the positive (rare) class alongside accuracy in Table 6. Sensitivity to this threshold ($q \in \{0.70, 0.80, 0.90\}$) is recorded in the JSON output for transparency. Binarisation discards information, so the resulting empirical τ and r should be read as lower bounds on the correlation lengths in the underlying continuous process.

Procedure

Empirical τ and r are computed using the same estimators as the synthetic experiments (Section 5). The three classical baselines are run with $M = 64$ on seed-permuted prequential orders (10 seeds, 95% confidence intervals). The proxy accuracy is acc_proxy $= \max(0.5, 1 - C^2 n \log(1/\delta)/T_{\mathrm{eff}})$ with $C = 2$, $\delta = 0.05$.

Findings (Tables 5 and 6)

The two streams land on opposite sides of the proxy-favourable region in (τ, r) space, and the imbalance-aware metrics expose effects that accuracy alone hides:

- **NYC Taxi** (favourable region): $\tau_{\mathrm{emp}} = 6.0$, $r_{\mathrm{emp}} = 2.77$, $T_{\mathrm{eff}} = 621$, proxy $= 0.884$. Count-Min reaches accuracy 0.865 ± 0.016, balanced accuracy 0.777, and $F_1 = 0.644$, materially above majority on every metric. Online-SGD and Averaged-SGD have lower accuracy (0.670 ± 0.006 and 0.673 ± 0.004) and balanced accuracy near 0.50, indicating they recover the rare class only weakly under hashed-linear features.
- **Machine Temperature** (unfavourable region): $\tau_{\mathrm{emp}} = 32.0$, $r_{\mathrm{emp}} = 13.55$, $T_{\mathrm{eff}} = 52$, proxy $= 0.500$ (saturated at the chance floor). All three classical baselines have accuracy near the majority value (0.724, 0.735, 0.801), but balanced accuracy is only 0.55, 0.54, and 0.56 and $F_1 \in [0.23, 0.27]$. The accuracy gap above the saturated proxy

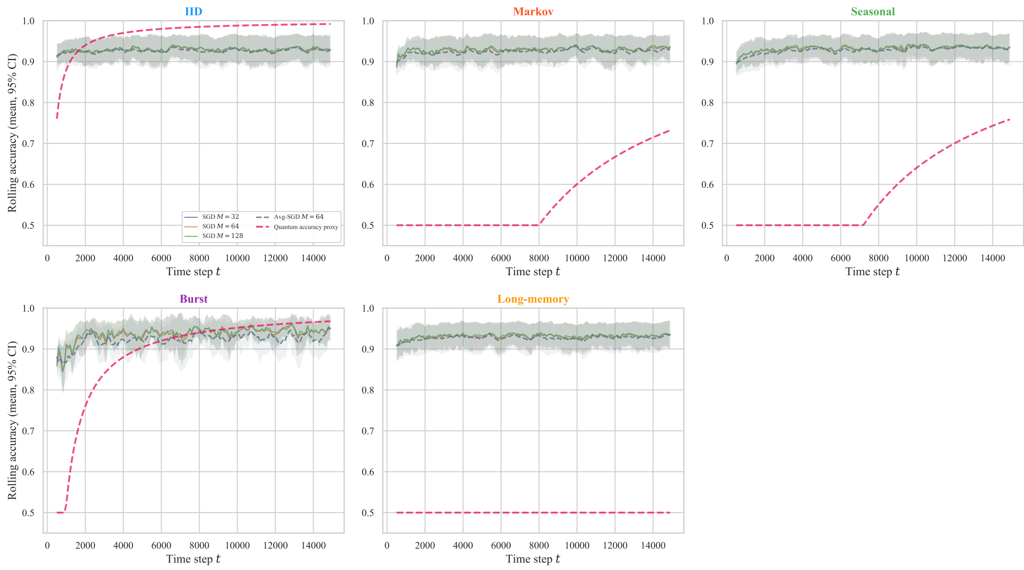

Figure 7. Rolling accuracy (window = 500) per regime; 10-seed mean ± 95% CI. Classical SGD at $M \in \{32, 64, 128\}$ (solid), averaged SGD $M = 64$ (dashed grey), quantum proxy (dashed pink). Under IID and burst the proxy reaches or exceeds classical by horizon end; under Markov and seasonal it rises more slowly and remains below classical (conservative analytic τ); under long-memory the proxy is pinned at 0.5 throughout.

Table 5. Real-stream sanity check, landscape coordinates: empirical (τ, r), proxy accuracy, classical-baseline accuracy (10-seed mean), and the majority-class accuracy for two NAB datasets. The proxy places NYC Taxi in a favourable region and Machine Temperature in an unfavourable region (proxy saturates at 0.5). Imbalance-aware metrics (balanced accuracy, F_1) for the same runs are reported separately in Table 6.

Dataset	T	τ	r	T_{eff}	proxy	SGD	Avg-SGD	Count-Min
NYC Taxi	10,320	6.0	2.77	621	0.884	0.670	0.673	0.865
Machine Temp.	22,695	32.0	13.55	52	0.500	0.724	0.735	0.801

is therefore largely majority-class memorisation; on the imbalance-aware metrics no classical baseline meaningfully exceeds chance for the rare class.

Why Count-Min looks strong on NYC Taxi

The near-proxy accuracy of Count-Min on NYC Taxi has a structural explanation that is worth stating up front, because otherwise the apparent agreement could be misread as evidence that the proxy is "correct". With $n = 6$ binary features only $2^n = 64$ feature vectors exist, and many of them repeat in the data; Count-Min can therefore essentially memorise per-bucket positive rates and predict the majority class for each bucket, which is enough to reach $F_1 = 0.644$ at this binarisation granularity. SGD with hashed-linear features cannot represent the same per-bucket conditional distribution as efficiently. The Count-Min result should be interpreted as a memorisation-friendly bounded-memory baseline under coarse binarisation, not as general evidence that the proxy predicts all classical learners. As n grows, 2^n outpaces practical Count-Min table sizes and this memorisation route disappears.

Interpretation

Three observations. (i) The empirical τ and r estimators apply unchanged to both real streams and place each stream in the regime that matches its underlying structure: NYC Taxi has moderate τ (multiple-bin daily/weekly correlation) and moderate r (modest exact-

Table 6. Imbalance-aware metrics for the real-stream sanity check (10-seed mean). Accuracy alone is misleading at a $\sim$20%/80% positive/negative split, so balanced accuracy and F_1 of the positive (rare) class are reported alongside it. The proxy is an accuracy projection only; balanced accuracy and F_1 are reported for the classical baselines and are not targets the proxy attempts to match. On NYC Taxi, Count-Min materially exceeds majority on every metric. On Machine Temperature, all classical baselines have accuracy slightly above majority but balanced accuracy near 0.5 and $F_1 < 0.30$, indicating the accuracy gap is largely majority-class memorisation.

Dataset	Method	accuracy	balanced acc.	F_1 (positive)
NYC Taxi	majority	0.800	0.500	0.000
	Online-SGD	0.670	0.535	0.273
	Averaged-SGD	0.673	0.498	0.199
	Count-Min	0.865	0.777	0.644
Machine Temperature	majority	0.800	0.500	0.000
	Online-SGD	0.724	0.550	0.269
	Averaged-SGD	0.735	0.538	0.234
	Count-Min	0.801	0.560	0.228

feature-vector repetition at $n = 6$); Machine Temperature has high τ (slow multi-hour drift) and very high r (5-min sensor data has many near-identical consecutive feature vectors). (ii) On NYC Taxi, Count-Min materially exceeds the majority baseline on balanced accuracy (0.777 vs. 0.500) and on F_1 (0.644 vs. 0.000), so the accuracy match between Count-Min and the proxy reflects genuine rare-class learning rather than majority memorisation alone, even though the underlying mechanism is the bounded-n memorisation route described above. (iii) On Machine Temperature, the saturated proxy is exceeded by all classical baselines on accuracy, but the imbalance-aware metrics show that the gap is mostly majority-class memorisation: balanced accuracy is at most 0.56 and F_1 is below 0.30 for every classical baseline. This is the expected diagnostic behaviour at $T_{\text{eff}} = 52$—no sample-limited estimator should be expected to recover the rare class at this effective sample count—and we treat it as a genuine negative case.

The two streams together illustrate the value of the (τ, r) diagnostic: it identifies real streams that fall on either side of the proxy-favourable boundary using the same estimators applied to the synthetic regimes, and the imbalance-aware metrics confirm that the diagnostic is not an artefact of accuracy on imbalanced labels. With only two datasets, however, this experiment is illustrative rather than statistically representative: it demonstrates that the diagnostic workflow runs end-to-end on real telemetry and produces interpretable placements, but it cannot establish that those placements generalise across stream domains. Broader real-stream coverage across financial, network-traffic, cybersecurity, and additional sensor domains remains future work (Section 9).

6.9. Experiment 9: Proxy-Sensitivity and Estimator Ablations

To address the concern that several proxy choices appear ad hoc, we vary five ingredients of the pipeline and report whether the qualitative ordering of regimes on the (τ, r) landscape is preserved. The first four ablations use $T = 4096$, $W = 20$, and a feature dimension $n = 6$ chosen to match the real-stream case study (Section 6.8) for direct comparability; the fifth (the C–δ analysis) re-evaluates the Markov sweep of Section 6.5 at its native $T = 10^4$, $n = 10$; the synthetic main experiments use $n = 10$, so absolute proxy values in this section are not directly comparable to those in Sections 6–6.4, but the qualitative regime ordering is.

Alternative τ estimators

Four definitions are compared: the default ($1/e$ ACF crossing), an aggressive variant (1/2 ACF crossing), the integrated absolute ACF up to first non-positive lag, and an AR(1)

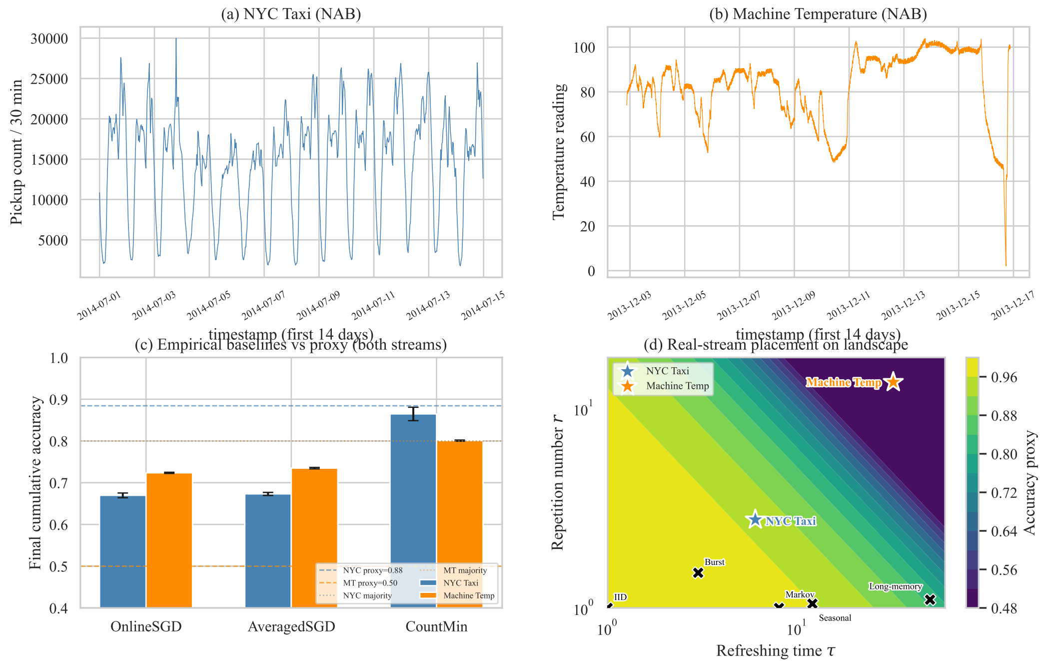

Figure 8. Real-stream sanity check (two NAB datasets). **(a, b)** Raw series for the first 14 days of each dataset. **(c)** Final cumulative classical baseline accuracy (10-seed mean $\pm$ 95% CI) for both streams; dashed lines are the deterministic proxy values, dotted lines are the majority baseline. **(d)** Both real-stream points (coloured stars) on the (τ, r) landscape, alongside synthetic-regime markers (black crosses). NYC Taxi falls in a moderately proxy-favourable region near the synthetic Markov point; Machine Temperature falls in the proxy-unfavourable upper-right region (high τ and very high r) where the proxy saturates. Imbalance-aware metrics for the same runs are reported in Table 6.

effective lag $-1/\log|\rho_1|$. The qualitative ordering is preserved across all five regimes: Burst always has the largest τ (3.3–7.0), IID and long-memory always the smallest (1.0–2.0), Markov consistently moderate (2.8–4.0). Resulting proxy accuracies (Table 8) shift by at most 0.04 between estimators; the largest shift is in the Burst regime, where the proxy is in saturation and the AR(1) estimator yields a modestly higher value because it captures only the lag-1 dependence.

Forward-window size W

Repeating the r estimator at $W \in \{10, 20, 50\}$ yields a monotone increase in r (more chances to find duplicates) and a corresponding monotone decrease in T_{eff}. The relative ordering of regimes by r is preserved across W values, confirming that $W = 20$ is a representative choice.

Stream length T

Varying $T \in \{1024, 2048, 4096, 8192\}$ yields the expected scaling in T_{eff}. For Burst the proxy emerges from saturation only at $T \geq 8192$, indicating sample-size sensitivity for highly repetitive regimes.

Target function

On the linear target the SGD baseline degrades to near-chance (≈ 0.5) for these synthetic streams and the small $n = 6$ feature set; on a threshold target ($y = 1$ iff $\sum_i x_i \geq n/2$) SGD attains ≈ 0.65 across all regimes; on the parity target SGD remains near chance. The proxy model is target-independent (it depends only on T_{eff}), so this ablation isolates the gap between the proxy framework and the empirical learner's task-specific capability. This is a scope boundary, stated plainly: classical learner performance is strongly target-dependent,

Table 7. Sensitivity of the proxy–baseline crossover ρ^* (Markov sweep, empirical τ, mean curves) to the proxy constants C and δ. Entries are interpolated on the 15-point sweep grid; "> 0.95" means the mean proxy curve remains above the mean classical curve over the entire sweep. Regime ordering is invariant across the grid; the absolute crossover is not. The paper's operating point ($C = 2$, $\delta = 0.05$) is set in bold.

	$\delta = 0.01$	$\delta = 0.05$	$\delta = 0.10$
$C = 1.0$	0.93	> 0.95	> 0.95
$C = 1.5$	0.84	0.89	0.92
$C = 2.0$	0.61	**0.82**	0.86
$C = 2.5$	0.38	0.61	0.76

Table 8. Proxy-sensitivity ablation: alternative τ estimators on synthetic streams ($T = 4096$, $W = 20$, $n = 6$). Qualitative ordering of regimes by τ is preserved across all four estimators; proxy accuracy changes by at most 0.04.

Regime	$1/e$ (default)	$1/2$-cross	integrated	AR(1) eff.
IID	0.954	0.954	0.977	0.977
Markov 0.7	0.856	0.892	0.877	0.899
Seasonal	0.942	0.942	0.893	0.971
Burst	0.500	0.500	0.500	0.655
Long-memory	0.954	0.954	0.977	0.977

while the proxy depends only on (τ, r, T) and is target-independent by construction. The proxy therefore cannot universally predict downstream classifier performance; it characterises the correlation structure of the stream—the effective sample budget available to *any* bounded-memory learner—not task-specific learnability. Predictions read off the landscape are statements about correlation-driven budget erosion and must not be interpreted as accuracy forecasts for a particular classifier–target pair (Section 7.4).

Proxy constants C and δ

Because the proxy is a closed-form function of the measured (τ, r), its sensitivity to the constants requires no new stochastic runs: we re-evaluate the accuracy proxy on the same per-seed empirical τ values from the Markov sweep (Section 6.5) over the grid $C \in \{1, 1.5, 2, 2.5\}$, $\delta \in \{0.01, 0.05, 0.1\}$, and record where the mean proxy curve crosses the mean classical curve. Table 7 shows that the regime *ordering* is invariant—as it must be, since C and δ enter only through the multiplicative factor $\kappa = C^2 \log(1/\delta)$, a monotone rescaling—but the *absolute* crossover location ρ^* moves substantially: from $\rho^* \approx 0.38$ at the most pessimistic setting (C=2.5, δ=0.01) to beyond the sweep range at $C = 1$. At the paper's operating point (C=2, δ=0.05), $\rho^* \approx 0.82$. The landscape should therefore be read *ordinally*: relative placements of regimes and streams are robust to the constants, while any absolute boundary—including the ρ^* values above—is conditional on the chosen (C, δ).

Together these ablations show that the proxy model's qualitative regime ordering is robust to τ estimator choice and forward-window size, while the absolute proxy values are sensitive to the constants C and δ (Table 7 quantifies the induced movement of the crossover ρ^*) and to the stream length T. The Burst regime is the most sensitive: at $T = 4096$ it is in proxy saturation under most estimators, but emerges from saturation as T doubles. These findings support the paper's claim that the (τ, r) landscape is a hypothesis-generating screening heuristic rather than a precise prediction tool.

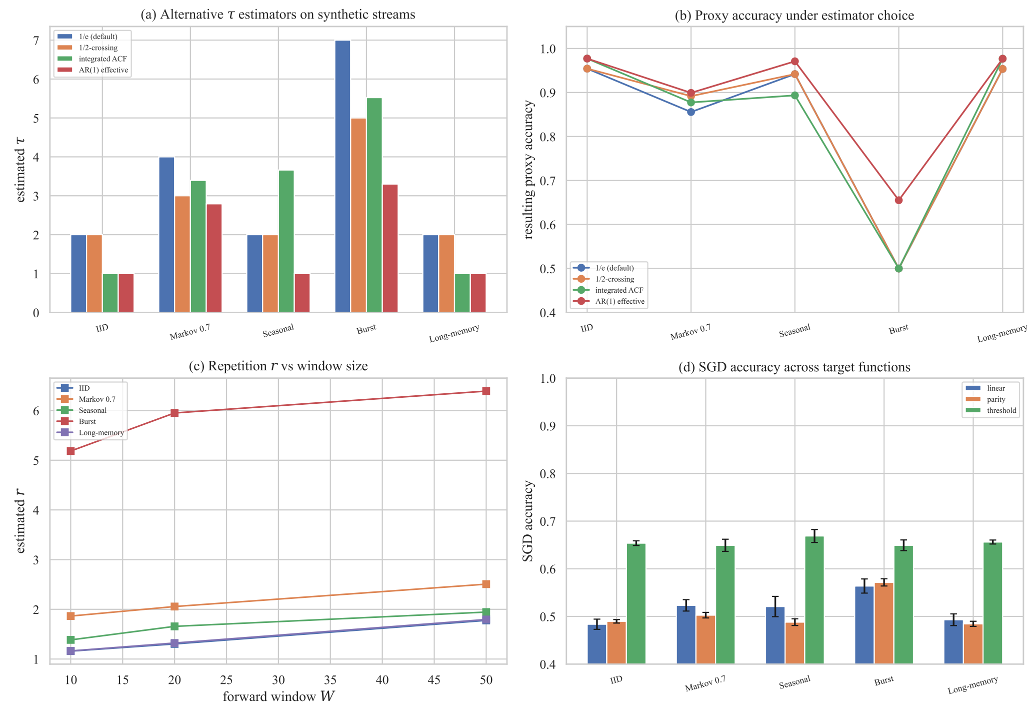

Figure 9. Proxy-sensitivity and estimator ablations. **(a)** Estimated τ on each regime under four alternative estimators. **(b)** Resulting proxy accuracy. **(c)** Repetition number r versus forward-window size W. **(d)** Online-SGD accuracy across three target functions (linear, parity, threshold), 10-seed mean with 95% CI. The proxy curve is target-independent by construction; it does not track task-specific learnability.

7. Discussion

7.1. Implications for Quantum Streaming Applications

Three structural factors are consistent with a hypothesized proxy-favourable regime: (1) the $\mathcal{O}(n)$ vs. $\Omega(\sqrt{N})$ worst-case memory-reference gap is a correlation-independent theoretical separation that grows exponentially with n, so at $n \geq 20$ even large increases in τ and r would not eliminate the proxy-side memory-separation reference (Section 3.3); (2) analytic proxy τ systematically overshoots empirical τ (Fig. 4; Section 7.3), so the proxy model is conservative and tighter family-specific characterizations would enlarge the predicted favourable region; (3) the proxy model predicts a favourable separation at $\rho \leq 0.47$ (effect sizes $r_{\mathrm{rb}} \geq +0.71$ with exact CIs excluding zero) that closes at $\rho^* \approx 0.82$–0.88 (Table 4), a range that overlaps autocorrelation levels reported in prior measurement studies; the two real-stream cases examined here (Section 6.8) place one stream inside and one outside the hypothesized favourable region, and broader validation across additional domains remains future work. All three factors are properties of the proxy model and the cited theory; none is a measured quantum result.

The (τ, r) landscape is a hypothesis-generating screening heuristic: researchers can place a candidate stream's estimated (τ, r) on the map to identify the most comparable synthetic regime and determine whether deeper empirical or algorithm-level study is warranted, with the contour shape robust to multiplicative proxy rescalings.

7.2. Practical Implications: Stream Classes on the Landscape

Table 9 translates typical autocorrelation ranges from the measurement literature into (τ, r) coordinates and records the proxy model's qualitative prediction. This table itself reports no new measurements; the overlays rely on published statistics [4,7,20,33,45,48,51]. The new real-stream measurements in this paper are the two NAB cases (Section 6.8): NYC Taxi falls in the favourable lower-left region of the landscape, while Machine Temperature

Table 9. Qualitative mapping of real stream classes onto the (τ, r) landscape (Fig. 3c). Reported parameter ranges are from the cited measurement literature. The final column records hypotheses generated by the proxy landscape for future algorithm-level testing, not demonstrated performance.

Stream class	Reported range	Landscape region	Proxy hypothesis
Network backbone	$H \approx 0.75$–0.95	upper-right (long-memory)	*Unfavourable*
Financial returns	$H \approx 0.55$–0.70	low–moderate τ, $r \approx 1$	*Favourable*
Sensor / IoT	burst + mid-ρ Markov	0.7–0.8 contour	*Borderline*
Concept-drifted	seasonal, moderate drift	0.7–0.8 contour (analytic τ)	*Modest at best*

falls in the unfavourable upper-right region where the proxy saturates. Running the full pipeline at scale—broader real-stream coverage paired with an implementable quantum oracle sketching algorithm—remains an explicit follow-on target (Section 9).

7.3. Why Analytic and Empirical Correlation Estimates Diverge

Experiment 4 found gaps of $\approx 8\times$ (Markov) and $\approx 125\times$ (long-memory) between analytic and empirical τ. These gaps are systematic, and understanding them matters for anyone using the landscape, so we give the technical explanation, its interpretation, and its implications.

Technical explanation

The two quantities estimate *different aspects* of temporal dependence. Each analytic proxy targets a model-level, worst-case dependence scale of the *generator*: for Markov switching it tracks the mixing time of the joint n-bit product chain ($\Theta(n \log n/(1-\rho))$, simplified to $n/(1-\rho)$), and for long-range dependence it tracks the growth of the integrated autocorrelation of the underlying Gaussian driver (T^{2H-1}), which is dominated by the heavy *tail* of the power-law autocorrelation function. The empirical estimator instead measures the first $1/e$ crossing of the *marginal, bitwise* autocorrelation of the *binarised* stream—an instance-level, short-lag quantity. For Markov switching, each individual bit decorrelates as ρ^k, crossing $1/e$ at $k \approx 1/\ln(1/\rho)$ (≈ 2.8 at $\rho = 0.7$), while the joint state of all n bits takes $\Theta(n \log n/(1-\rho))$ steps to mix; both numbers are correct answers to different questions. For long-range dependence the divergence is compounded three ways: the underlying process has no finite decorrelation time at all (its autocorrelation is non-summable), so any finite empirical τ is a property of the estimator rather than of the process; binarisation shrinks the autocorrelation via the arcsine law $(2/\pi)\arcsin(\gamma)$ [7]; and the $1/e$-crossing responds to the fast initial decay of the power-law autocorrelation while the analytic proxy integrates its slow tail.

Interpretation

The divergence is partly *expected behaviour* and partly a *proxy limitation*, and the long-memory regime shows why the two cannot be fully separated: for a non-mixing process there is no single scalar that both estimators could agree on, so a single-number τ is intrinsically lossy there. It is *not* evidence of a data-generation defect: the fGn generator reproduces the intended Hurst statistics, and the gap persists under all four alternative τ estimators in Experiment 9, which shows that the estimator family, not an implementation artefact, is responsible.

Implications

First, the empirical τ is the operationally meaningful input for placing a stream on the landscape: it reflects the short-lag effective information that a bounded-memory learner actually experiences over the measured horizon. Second, the analytic τ retains value as a conservative stress-test envelope: a stream that remains in the hypothesized favourable region even under the analytic τ is favourable with margin. Third, the *ratio* of analytic to empirical τ is itself a useful diagnostic—large ratios flag long-memory content whose risk a short-lag estimator will underestimate at longer horizons. Multi-scale refinements of τ (e.g., a family of integrated-autocorrelation estimates at several horizons) are a natural follow-on (Section 9).

7.4. Scope of the Diagnostic: What the Landscape Can and Cannot Predict

The experiments delimit the diagnostic sharply, and we state the boundary explicitly. The landscape *can*: (i) rank correlation regimes and real streams by proxy-estimated effective sample budget, using estimators that apply unchanged to synthetic and real streams (Experiments 4 and 8); (ii) flag streams whose correlation structure erodes the budget so severely that no sample-limited method should be expected to recover a rare class (Machine Temperature at $T_{\text{eff}} = 52$); and (iii) generate falsifiable, regime-level hypotheses about where an implemented oracle-sketching algorithm would be most worth evaluating. The landscape *cannot*: (i) predict the accuracy of a specific classifier on a specific target—classical performance is target-dependent while the proxy is target-independent (Experiment 9, Fig. 9d); (ii) anticipate representation-specific effects such as the Count-Min memorisation route on coarse binarisations (Experiment 8); or (iii) certify quantum performance, because no quantum algorithm is implemented. In short, the landscape characterises correlation structure, not task-specific learning performance, and its outputs are hypotheses for validation rather than performance guarantees.

8. Threats to Validity

We systematise the study's limitations as threats to validity, stating for each how it is mitigated and what residual risk remains.

Construct validity

The central constructs—τ, r, T_{eff}, and the accuracy proxy—are operational stand-ins for theoretical quantities they do not provably equal (Section 3.3). Four specific risks follow. (i) No quantum algorithm is implemented or simulated, so "quantum" curves measure the heuristic model, not quantum computation. (ii) τ and r are not formally equivalent to the exact framework's refreshing time and repetition number, and Section 7.3 shows that a single-scalar τ is intrinsically lossy for non-mixing processes. (iii) The accuracy proxy applies a task-level generalisation rate rather than the oracle-construction cost, and the two scale differently in n (Experiment 6). (iv) Feature binarisation changes the correlation structure it is meant to measure (arcsine shrinkage), so real-stream (τ, r) estimates are lower bounds on the underlying continuous-process correlation. Mitigations: the provenance of every quantity is tabulated (Table 1), and the estimator ablations (Experiment 9) bound the sensitivity of conclusions to the estimator family.

Internal validity

(i) The headline comparisons pair a stochastic learner with a deterministic-given-$\hat{\tau}$ model prediction; they are seed-stability statements, not algorithm-versus-algorithm outcomes, and are labelled as such (Experiment 5). (ii) The same stream realisation is used both to estimate (τ, r) and to evaluate the classical baselines, which could couple estimation error with baseline noise; the narrow CI bands and the estimator ablations bound this effect. (iii) Classical hyperparameters (learning rate, hash width M) are fixed across regimes rather than tuned per regime; conclusions about the proxy are unaffected, but per-regime classical accuracies should not be read as the best classically achievable. (iv) The classical baselines are practical learners on a simple linear task, not the worst-case hard instances of the cited lower bounds, so no comparison here is an algorithm-versus-algorithm benchmark of the theory.

External validity

(i) The five synthetic families are author-chosen idealisations of documented stream phenomena; other correlation structures (regime-switching with heavy-tailed sojourns, adversarial drift) are unrepresented. (ii) Only two real-world datasets are analysed, both from NAB telemetry; the real-stream findings are illustrative, and generalisation to network telemetry, cybersecurity event streams, financial series, and large IoT fleets is unverified (Section 9). (iii) Dimensions are modest ($n \leq 14$), while the theoretically compelling memory separations concern $n \geq 20$; moreover $\tau_{\text{Markov}} = n/(1-\rho)$ grows linearly in n, so the analytic Markov proxy becomes less favourable exactly where the memory argument strengthens, making family-specific refinements most valuable there. (iv) The rare-event target on real streams is a per-stream 80th-percentile threshold, not the original NAB anomaly labels, so results do not transfer to NAB benchmark comparisons.

Statistical-conclusion validity

All stochastic results use 10 seeds with 95% CIs, but $n = 10$ limits power: the five sweep comparisons do not survive Holm correction at the family-wise 0.05 level, and the crossover claim therefore rests on effect sizes and interval estimates rather than significance (Table 4). The paired tests' interpretation is further restricted by the deterministic-vs-stochastic pairing above. Absolute landscape boundaries are conditional on the constants (C, δ) (Table 7); only ordinal statements are robust across the grid.

9. Future Work

The framework's outputs are hypotheses, and the priority for future work is to test them. (i) *Algorithm-level validation*: implement or simulate an oracle-sketching pipeline at small n and compare its measured behaviour against the landscape's regime-level predictions—the single step that would convert the hypotheses generated here into evidence. (ii) *Formal bridging*: prove equivalence of, or bounds between, the operational (τ, r) and the exact framework's correlation parameters, including multi-scale refinements of τ for non-mixing processes (Section 7.3). (iii) *Broader real-stream corpora*: extend the two-dataset sanity check to network telemetry (flow and latency series), cybersecurity event streams (intrusion and log-anomaly feeds), financial tick and returns series, and large-scale IoT monitoring fleets, characterising each stream's (τ, r) and mapping it onto Fig. 3 with imbalance-aware metrics throughout. (iv) *Tighter family-specific proxies* that close the analytic–empirical τ gap. (v) *Task extensions* to multiclass classification and regression, and repetition-aware oracle constructions that exploit burst structure variance-reductively rather than treating duplicates as wasted samples.

10. Conclusion

This paper introduced a proxy-based diagnostic and hypothesis-generation framework for correlation-sensitive streaming analysis, and applied it in a computational study spanning five structured synthetic non-IID regimes, two real telemetry streams, and a five-way ablation suite. The **primary contribution** is the (τ, r) sensitivity landscape: an operational map, built from measurable correlation proxies, on which synthetic regimes and real streams are placed with the same estimators. The **secondary contribution** is the use of that landscape for hypothesis generation: it identifies where quantum oracle sketching is most likely to remain robust, and therefore where algorithm-level validation effort should be spent first. The paper demonstrates no quantum advantage and implements no quantum algorithm; every quantum-side quantity is a heuristic proxy whose provenance and epistemic status are tabulated in Section 3.3.

Research question answers

- **RQ1**: Analytic proxies overestimate empirical τ by $\approx 8\times$ (Markov) to $\approx 125\times$ (long-memory); the generic robustness proxy is qualitatively informative but quantitatively loose, and Section 7.3 explains the divergence as two estimators answering different questions about temporal dependence.
- **RQ2**: The (τ, r) landscape (Fig. 3) organises all five regimes and both real streams consistently, and its qualitative ordering is robust to estimator choice, window size,

and the proxy constants (Experiment 9)—though absolute boundaries are conditional on (C, δ) (Table 7).
- **RQ3**: Under the empirical-τ proxy model, the hypothesized favourable region covers IID, burst, and mild-to-moderate Markov correlation; the paired proxy − classical difference shrinks monotonically (Hodges–Lehmann $+0.042 \rightarrow +0.019$) and reverses sign at $\rho^* \approx 0.82$–0.88 (-0.046, exact CI excluding zero); the proxy-predicted separation vanishes entirely in the long-memory regime.
- **RQ4**: Analytic r and empirical r agree within $1.4\times$ for burst and are near-identical for all other regimes under unit-aligned measurement; the directional τ gaps are monotone in correlation strength and admit family-specific refinement.

Practical contribution

For stream-processing practitioners, the framework is a cheap pre-screening step: estimate (τ, r) with the paper's estimators, place the stream on the landscape, and read off whether correlation alone already erodes the effective sample budget below usefulness—as it does for the Machine Temperature stream ($T_{\text{eff}} = 52$)—before investing in any advanced pipeline, quantum or classical. The real-stream check (Experiment 8) shows the workflow runs end-to-end on public telemetry, with imbalance-aware metrics guarding against majority-class artefacts.

Limitations and required validation

The framework's outputs are hypotheses conditioned on a heuristic model; Section 8 systematises the construct, internal, external, and statistical threats. The decisive missing step is algorithm-level validation: an implemented or simulated oracle-sketching pipeline evaluated at the landscape's predicted-favourable and predicted-unfavourable coordinates (Section 9). Until that step is taken, the landscape should be used as a prioritisation tool, not as evidence of quantum benefit.

Author Contributions: Conceptualization, A.H.; methodology, A.H. and M.M.; software, M.M.; validation, M.F.; formal analysis, M.M.; investigation, M.M.; visualization, M.M.; writing—original draft preparation, M.M.; writing—review and editing, A.H. and M.F.; supervision, A.H. All authors have read and agreed to the published version of the manuscript.

Funding: This research received no external funding.

Institutional Review Board Statement: Not applicable.

Informed Consent Statement: Not applicable.

Data Availability Statement: The two real-world datasets analysed in this work—NYC Taxi pickup counts and machine-temperature sensor readings—are publicly available under the MIT licence in the Numenta Anomaly Benchmark repository [2]. All other data are synthetic and generated by seeded random-number generators with the parameters specified in Section 5 and Table 3; every stochastic experiment uses 10 seeds (indices 0–9). A reference Python implementation of the stream generators, classical baselines, proxy formulas, real-stream loader, and ablation harness is openly available under the MIT licence and archived at Zenodo (DOI `10.5281/zenodo.19831893`), with an active mirror on GitHub (https://github.com/mmahmoudai/Beyond-IID-Data-A-Proxy-Based-Computational-Study-of-Quantum-Oracle-Sketchin). The repository regenerates every figure and table reported in this paper.

Acknowledgments: During the preparation of this manuscript the authors used Claude (Anthropic) to assist with drafting, language editing, and formatting; the authors reviewed, verified, and edited all content and take full responsibility for the content of this publication.

Conflicts of Interest: The authors declare no conflicts of interest.

References

1. Aaronson, S., 2015. Read the fine print. *Nature Physics* 11 (4), 291–293. https://doi.org/10.1038/nphys3272.

2. Ahmad, S., Lavin, A., Purdy, S., Agha, Z., 2017. Unsupervised real-time anomaly detection for streaming data. *Neurocomputing* 262, 134–147. https://doi.org/10.1016/j.neucom.2017.04.070.
3. Alon, N., Matias, Y., Szegedy, M., 1999. The space complexity of approximating the frequency moments. *Journal of Computer and System Sciences* 58 (1), 137–147. https://doi.org/10.1006/jcss.1997.1545.
4. Arora, S., Rani, R., Saxena, N., 2024. A systematic review on detection and adaptation of concept drift in streaming data using machine learning techniques. *WIREs Data Mining and Knowledge Discovery* 14 (4), e1536. https://doi.org/10.1002/widm.1536.
5. Arunachalam, S., de Wolf, R., 2018. Optimal quantum sample complexity of learning algorithms. *Journal of Machine Learning Research* 19 (71), 1–36. https://jmlr.org/papers/v19/18-195.html.
6. Ben Mazziane, Y., Alouf, S., Neglia, G., 2022. Analyzing count-min sketch with conservative updates. *Computer Networks* 217, 109315. https://doi.org/10.1016/j.comnet.2022.109315.
7. Beran, J., 1994. *Statistics for Long-Memory Processes*. Chapman & Hall/CRC, Boca Raton. ISBN 978-0412049019. https://doi.org/10.1201/9780203738481.
8. Ben-David, S., Blitzer, J., Crammer, K., Kulesza, A., Pereira, F., Vaughan, J.W., 2010. A theory of learning from different domains. *Machine Learning* 79, 151–175. https://doi.org/10.1007/s10994-009-5152-4.
9. Besbes, O., Gur, Y., Zeevi, A., 2015. Non-stationary stochastic optimization. *Operations Research* 63 (5), 1227–1244. https://doi.org/10.1287/opre.2015.1408.
10. Biamonte, J., Wittek, P., Pancotti, N., Rebentrost, P., Wiebe, N., Lloyd, S., 2017. Quantum machine learning. *Nature* 549 (7671), 195–202. https://doi.org/10.1038/nature23474.
11. Bifet, A., Gavaldà, R., 2007. Learning from time-changing data with adaptive windowing. In *Proceedings of the 2007 SIAM International Conference on Data Mining*, pp. 443–448. https://doi.org/10.1137/1.9781611972771.42.
12. Bifet, A., Holmes, G., Kirkby, R., Pfahringer, B., 2010. MOA: massive online analysis. *Journal of Machine Learning Research* 11, 1601–1604. https://jmlr.org/papers/v11/bifet10a.html.
13. Cerezo, M., Arrasmith, A., Babbush, R., Benjamin, S.C., Endo, S., Fujii, K., McClean, J.R., Mitarai, K., Yuan, X., Cincio, L., Coles, P.J., 2021. Variational quantum algorithms. *Nature Reviews Physics* 3 (9), 625–644. https://doi.org/10.1038/s42254-021-00348-9.
14. Chakraborty, S., Gilyén, A., Jeffery, S., 2019. The power of block-encoded matrix powers: improved regression techniques via faster Hamiltonian simulation. In *Proceedings of the 46th International Colloquium on Automata, Languages, and Programming (ICALP)*, Article 33. https://doi.org/10.4230/LIPIcs.ICALP.2019.33.
15. Chen, L., Li, T., Chen, Y., Chen, X., Woźniak, M., Xiong, N., Liang, W., 2024. Design and analysis of quantum machine learning: a survey. *Connection Science* 36 (1), 2312121. https://doi.org/10.1080/09540091.2024.2312121.
16. Cormode, G., Muthukrishnan, S., 2005. An improved data stream summary: the count-min sketch and its applications. *Journal of Algorithms* 55 (1), 58–75. https://doi.org/10.1016/j.jalgor.2003.12.001.
17. Cormode, G., Garofalakis, M., Haas, P.J., Jermaine, C., 2011. Synopses for massive data: samples, histograms, wavelets, sketches. *Foundations and Trends in Databases* 4 (1–3), 1–294. https://doi.org/10.1561/1900000004.
18. Ditzler, G., Roveri, M., Alippi, C., Polikar, R., 2015. Learning in nonstationary environments: a survey. *IEEE Computational Intelligence Magazine* 10 (4), 12–25. https://doi.org/10.1109/MCI.2015.2471196.
19. Gama, J., 2010. *Knowledge Discovery from Data Streams*. Chapman & Hall/CRC, Boca Raton. https://doi.org/10.1201/EBK1439826119.
20. Gama, J., Žliobaitė, I., Bifet, A., Pechenizkiy, M., Bouchachia, A., 2014. A survey on concept drift adaptation. *ACM Computing Surveys* 46 (4), Article 44. https://doi.org/10.1145/2523813.
21. Ghashami, M., Liberty, E., Phillips, J.M., Woodruff, D.P., 2016. Frequent directions: simple and deterministic matrix sketching. *SIAM Journal on Computing* 45 (5), 1762–1792. https://doi.org/10.1137/15M1009718.
22. Gilyén, A., Su, Y., Low, G.H., Wiebe, N., 2019. Quantum singular value transformation and beyond: exponential improvements for quantum matrix arithmetics. In *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing*, pp. 193–204. https://doi.org/10.1145/3313276.3316366.
23. Giovannetti, V., Lloyd, S., Maccone, L., 2008. Quantum random access memory. *Physical Review Letters* 100 (16), 160501. https://doi.org/10.1103/PhysRevLett.100.160501.
24. Gomes, H.M., Bifet, A., Read, J., Barddal, J.P., Enembreck, F., Pfahringer, B., Holmes, G., Abdessalem, T., 2017. Adaptive random forests for evolving data stream classification. *Machine Learning* 106 (9–10), 1469–1495. https://doi.org/10.1007/s10994-017-5642-8.
25. Harrow, A.W., Hassidim, A., Lloyd, S., 2009. Quantum algorithm for linear systems of equations. *Physical Review Letters* 103 (15), 150502. https://doi.org/10.1103/PhysRevLett.103.150502.
26. Havlíček, V., Córcoles, A.D., Temme, K., Harrow, A.W., Kandala, A., Chow, J.M., Gambetta, J.M., 2019. Supervised learning with quantum-enhanced feature spaces. *Nature* 567 (7747), 209–212. https://doi.org/10.1038/s41586-019-0980-2.
27. Hoi, S.C.H., Sahoo, D., Lu, J., Zhao, P., 2021. Online learning: a comprehensive survey. *Neurocomputing* 459, 249–289. https://doi.org/10.1016/j.neucom.2021.04.112.
28. Huang, H.-Y., Kueng, R., Preskill, J., 2020. Predicting many properties of a quantum system from very few measurements. *Nature Physics* 16 (10), 1050–1057. https://doi.org/10.1038/s41567-020-0932-7.
29. Huang, H.-Y., Broughton, M., Mohseni, M., Babbush, R., Boixo, S., Neven, H., McClean, J.R., 2021. Power of data in quantum machine learning. *Nature Communications* 12, 2631. https://doi.org/10.1038/s41467-021-22539-9.

30. Huang, H.-Y., Broughton, M., Cotler, J., Chen, S., Li, J., Mohseni, M., Neven, H., Babbush, R., Kueng, R., Preskill, J., McClean, J.R., 2022. Quantum advantage in learning from experiments. *Science* 376 (6598), 1182–1186. https://doi.org/10.1126/science.abn7293.
31. Kallaugher, J., 2021. A quantum advantage for a natural streaming problem. In *Proceedings of the 62nd IEEE Annual Symposium on Foundations of Computer Science (FOCS)*, pp. 897–908. https://doi.org/10.1109/FOCS52979.2021.00091.
32. Kallaugher, J., Parekh, O., Voronova, N., 2025. How to design a quantum streaming algorithm without knowing anything about quantum computing. In *Proceedings of the 2025 Symposium on Simplicity in Algorithms (SOSA)*, pp. 9–45. https://arxiv.org/abs/2410.18922.
33. Leland, W.E., Taqqu, M.S., Willinger, W., Wilson, D.V., 1994. On the self-similar nature of Ethernet traffic (extended version). *IEEE/ACM Transactions on Networking* 2 (1), 1–15. https://doi.org/10.1109/90.282603.
34. Levin, D.A., Peres, Y., 2017. *Markov Chains and Mixing Times*, 2nd ed. American Mathematical Society, Providence, RI. ISBN 978-1470429621. https://doi.org/10.1090/mbk/107.
35. Lloyd, S., Mohseni, M., Rebentrost, P., 2014. Quantum principal component analysis. *Nature Physics* 10 (9), 631–633. https://doi.org/10.1038/nphys3029.
36. Mandelbrot, B.B., Van Ness, J.W., 1968. Fractional Brownian motions, fractional noises and applications. *SIAM Review* 10 (4), 422–437. https://doi.org/10.1137/1010093.
37. Muthukrishnan, S., 2005. Data streams: algorithms and applications. *Foundations and Trends in Theoretical Computer Science* 1 (2), 117–236. https://doi.org/10.1561/0400000002.
38. Park, K., Willinger, W. (Eds.), 2000. *Self-Similar Network Traffic and Performance Evaluation*. Wiley-Interscience, New York. https://doi.org/10.1002/047120644X.
39. Polyak, B.T., Juditsky, A.B., 1992. Acceleration of stochastic approximation by averaging. *SIAM Journal on Control and Optimization* 30 (4), 838–855. https://doi.org/10.1137/0330046.
40. Preskill, J., 2018. Quantum computing in the NISQ era and beyond. *Quantum* 2, 79. https://doi.org/10.22331/q-2018-08-06-79.
41. Rebentrost, P., Mohseni, M., Lloyd, S., 2014. Quantum support vector machine for big data classification. *Physical Review Letters* 113 (13), 130503. https://doi.org/10.1103/PhysRevLett.113.130503.
42. Schuld, M., Petruccione, F., 2018. *Supervised Learning with Quantum Computers*. Quantum Science and Technology. Springer, Cham. https://doi.org/10.1007/978-3-319-96424-9.
43. Schuld, M., Sweke, R., Meyer, J.J., 2021. Effect of data encoding on the expressive power of variational quantum-machine-learning models. *Physical Review A* 103 (3), 032430. https://doi.org/10.1103/PhysRevA.103.032430.
44. Shalev-Shwartz, S., 2012. Online learning and online convex optimization. *Foundations and Trends in Machine Learning* 4 (2), 107–194. https://doi.org/10.1561/2200000018.
45. Su, R., Guo, H., Wang, W., 2024. Elastic online deep learning for dynamic streaming data. *Information Sciences* 676, 120783. https://doi.org/10.1016/j.ins.2024.120783.
46. Tang, E., 2019. A quantum-inspired classical algorithm for recommendation systems. In *Proceedings of the 51st Annual ACM SIGACT Symposium on Theory of Computing*, pp. 217–228. https://doi.org/10.1145/3313276.3316310.
47. Weinberger, K., Dasgupta, A., Langford, J., Smola, A., Attenberg, J., 2009. Feature hashing for large scale multitask learning. In *Proceedings of the 26th International Conference on Machine Learning (ICML)*, pp. 1113–1120. https://doi.org/10.1145/1553374.1553516.
48. Willinger, W., Taqqu, M.S., Sherman, R., Wilson, D.V., 1997. Self-similarity through high-variability: statistical analysis of Ethernet LAN traffic at the source level. *IEEE/ACM Transactions on Networking* 5 (1), 71–86. https://doi.org/10.1109/90.554723.
49. Woodruff, D.P., 2014. Sketching as a tool for numerical linear algebra. *Foundations and Trends in Theoretical Computer Science* 10 (1–2), 1–157. https://doi.org/10.1561/0400000060.
50. Wu, Z., Wang, H., Guo, J., Yang, Q., Shao, J., 2024. Learning evolving prototypes for imbalanced data stream classification with limited labels. *Information Sciences* 679, 120979. https://doi.org/10.1016/j.ins.2024.120979.
51. Yan, H., Liu, J., Xiao, J., Niu, S., Dong, S., You, D., Shen, L., 2024. Online learning for data streams with bi-dynamic distributions. *Information Sciences* 676, 120796. https://doi.org/10.1016/j.ins.2024.120796.
52. Zhao, H., Zlokapa, A., Neven, H., Babbush, R., Preskill, J., McClean, J.R., Huang, H.-Y., 2026. Exponential quantum advantage in processing massive classical data. arXiv:2604.07639. Preprint. https://arxiv.org/abs/2604.07639.